

Yixuan Yang

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EDUCATION

Zhejiang University	Hangzhou, China
Ph.D Candidate, Applied Mathematics	2022.09 - Present
School of Mathematical Sciences	
Advisor: Prof. Song Li.	

Lanzhou University	Lanzhou, China
Bachelor of Science, Pure and Applied Mathematics	2018.09 - 2022.06
School of Mathematics and Statistics	

RESEARCH INTERESTS

My research primarily focuses on mathematical foundations of Data Science, specifically in deep learning theory and non-smooth optimization. I am also interested in theory, algorithm and application of transformer and Large Language Model.

PUBLICATIONS & PREPRINTS

1. Yixuan Yang, Yuqing He, Song Li, Convergence of Spectral Descent for Non-smooth Optimization. arXiv preprint, 2026.
2. Xiaoxuan Ma, Yixuan Yang, Song Li, Xiangyun Hui, How Label Imbalance Shapes Geometry: A General Spectral Analysis of Multi-Label Neural Collapse. arXiv preprint, 2026.
3. Xiangyun Hui, Xiaoxuan Ma, Yixuan Yang, Song Li, The implicit regularization of gradient flow on separable datasets in ReLU networks. Neurocomputing, Volume 656, December 2025.

TEACHING EXPERIENCE

Zhejiang University	Hangzhou, China
Teaching assistant: Probability and Mathematical Statistics	Spring 2024, 2025, 2026; Fall 2023, 2025
Teaching assistant: Ordinary Differential Equations	Spring 2023
Teaching assistant: Mathematical Analysis (III)	Fall 2024

SELECTED AWARDS & HONOURS

Five-Excellence Graduate Student of ZJU	2023, 2025
Outstanding Graduate of LZU	2022
National Scholarship	2021, 2020